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TiLiA – TimeLineAnnotator

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Summary

'TiLiA' (**Ti**me**Li**ne **A**nnotator) is an open-source, cross-platform network of tools and resources designed to facilitate real-time creating, displaying and interacting with timeline-style annotations of audio, video and musical scores. Developed in Python with the PySide UI framework, TiLiA is available primarily as a desktop Graphical User Interface (GUI), supported by a Command Line Interface (CLI) as well as a free and open online platform where users may upload, view and query existing analyses.

Statement of need

Visual analyses of music have proven to be an enduringly popular, effective, and versatile tool for conveying a variety of musical ideas.[1] As the famous adage goes: 'a picture is worth a thousand words'. A comprehensive analysis of music requires being able to summarise the relationships between events that are happening concurrently and through time. Furthermore, whatever potential a *static* image has, an *interactive* one offers considerably more. Form analysis, for instance, can be hard to parse (especially in real-time) but is often easy to understand through visual summaries.

In today's digital age, we have come to expect certain basic interactive features across visual interfaces, such as resizing content to fit the window, keyboard shortcuts, undo/redo functionality, and so on. Musical contexts extend this with additional expectations, such as real-time synchronisation (a.k.a., audio or score following).

Despite several emergent initiatives (see below), musical scholarship has not yet made the leap into this technological space with any degree of decisiveness or consistency. Even most visually minded music analyses are consigned to static images embedded in print (or in e-copies at best) and not subject to the expectations for *fair*, open-source data sharing that are now *de rigueur* in other fields.

State of the field

While one *can* appropriate score and audio editors for visual analysis (Peebles, 2013), they are limited in their annotating capabilities. Dedicated tools fall into a few groups, according to their focus. Audio-analysis tools such as Sonic Visualiser (Cannam et al., 2010) and Partiels (Guillot, 2025) centre on audio and plug-in features, and lack score integration. Score-annotation platforms such as *Edirom Online* (Röwenstrunk, 2014), *Listen Here!* (Weigl et al., 2023) and *mei-friend* (Goebl & Weigl, 2024) anchor annotations to notated elements rather than to time and lack dedicated types for analytical structures such as form. "Timeline-based" annotators such as *Audio Timeliner* (Notess & Swan, 2004; Yorgason, 2018), *iAnalyse* (Couprie, 2019),

[1] See Isaacson (2023) for a survey of this long history.

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BriFormer (Jarvis, 2020) and *Dezrann* (Giraud et al., 2018) tend to focus more on providing tools for rich hand-made analysis.[2] While each group excels at particular tasks, there remains space for a solution that caters to more cross-cutting needs.

Availability, for instance, is often constrained (Table 1): iAnalyse runs only on macOS, while Dezrann and BriFormer are hosted web applications. BriFormer has no open licence and depends on its author's website. Dezrann places uploaded pieces in a public sandbox; working with a private recording means requesting an account by e-mail or deploying the platform oneself. Few of the tools are scriptable, and moving analyses between tools can be cumbersome, constraining interoperability and work at scale.

Most software focuses on a subset of the many forms analytical annotations can take (e.g. hierarchical groupings, metre, overlapping time spans, score elements, harmony, text). Audio Timeliner and BriFormer allow only nested spans, whereas Sonic Visualiser and iAnalyse allow overlaps, but only on a flat surface. Datasets need all of them: SALAMI (Smith et al., 2011) and the Wagner Ring Dataset (Weiß et al., 2023) encode hierarchical structure, a topic of active research (McFee & Ellis, 2014); the ABC (Neuwirth et al., 2018) and the Annotated Mozart Sonatas (Hentschel et al., 2021) layer harmony, phrase and cadence labels; BPSD (Zeitler et al., 2024) and the Harmonix Set (Nieto et al., 2019) align formal structure with beats and measures; and the Schubert Winterreise Dataset (Weiß et al., 2021) links harmony annotations to scores and several recordings.

Input formats and multimodality have varied support. Symbolic-based platforms, for instance, rarely support video. The Walküre (Balke et al., 2017), Freischütz Digital (Müller et al., 2013), Erkomaishvili (Rosenzweig et al., 2020) and Lohengrin TimeMachine (Lewis et al., 2021) interfaces show what multimodal, synchronised and richly interactive access can offer, but are tied to specific corpora.

Table 1: Comparison of music annotation tools.

Tool	Annotation type							Media					Licence	Script/Platform				
	B	C	E	H	I	O	T	A	P	S	V	Y		IO	L	M	W	w
TiLiA	●	●	○	●	●	●		●	●	●	●	●	GPL-3.0	CLI; CSV, JSON	●	●	●	○
Audio Timeliner				●	●		●	●					–	CLI		●	●	
BriFormer				●	●	○		●				●	–	–				●
Dezrann		●	●	○	●	●		●	●	●		●	GPL-3.0	JSON				●
iAnalyse			●		●	●	●	●	●	●	●		–	–		●		
mei-friend		●	●		●		●		●	●			AGPL-3.0	MEI				●
Sonic Visualiser	●	●			●	●	●	●					GPL-2.0	CSV, CLI	●	●	●	

Annotations: **B** beat · **C** harmony · **E** score elements · **H** hierarchical (nested, e.g. phrases within sections) · **I** instant (single timepoints, e.g. cadences) · **O** overlapping (spans that may coincide, e.g. motifs) · **T** text
Media: **A** audio · **P** PDF · **S** score · **V** video · **Y** YouTube
Platforms: **L** Linux · **M** macOS · **W** Windows · **w** web
Hollow circles mark partial support.

[2] Commercial alternatives include: *Hookpad* and *Soundslice* (closed-source and subscription-based). SyncPlayer (Fremerey et al., 2007; Kurth et al., 2005) was another timeline-based annotator that is no longer available.

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In response to this challenge, we present 'TiLiA': a timeline annotator. It provides an open-source, cross-platform desktop tool that aligns annotations with audio, video, YouTube, PDFs and scores on shared timelines, offers dedicated types for musical structure, harmony and metre, and can be scripted and exchanged through CSV and JSON.

Specifications

TiLiA consists of an organisation of digital tools, primarily of its open-source desktop application and command-line interface, but also of a supporting online platform. The desktop application is developed in Python using the PySide binding for the Qt UI framework. Automated testing is done with `pytest`, and fully-automated deployment is available via GitHub Actions. The code base is loosely organised around a Model-View-Controller (MVC) pattern, which allows both the CLI and the GUI to benefit from the same backend logic. We provide object-oriented base classes such as `Timeline`, `TimelineUI` and `TimelineComponent` as a means to promote extensibility and support an evergrowing range of annotation needs.

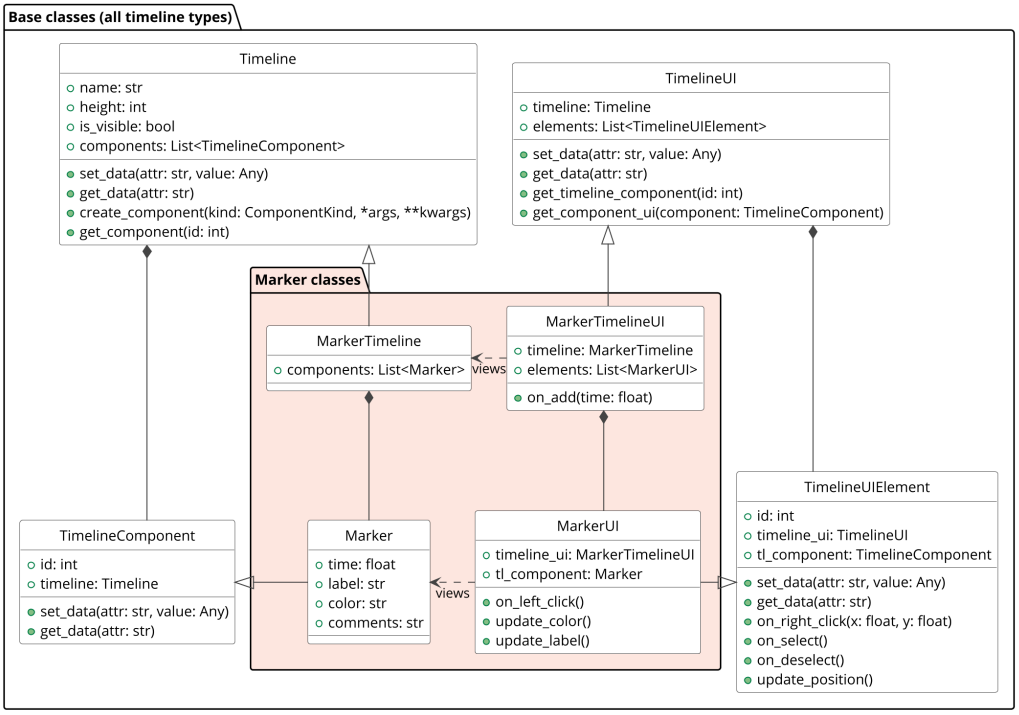


Figure 1: A *simplified* diagram for the main timeline base classes.

Figure 1 is a *simplified* diagram for the main timeline base classes. The outer box shows the base classes shared by all timeline types; the inner box shows the concrete subclasses for the marker timeline type, one example among the various timeline types. `Timeline` acts as container for `TimelineComponents`. Both act as models of `TimelineUI` and `TimelineUIElement`, which are views; in the marker example, `MarkerTimelineUI` and `MarkerUI` view `MarkerTimeline` and `Marker` respectively.

TiLiA is designed both for the manual creation of analyses — particularly where this requires complex alignment with audio/video sources — and for the large-scale conversion and import of external analyses. While the `.tla` format may still change as functionality grows, documentation is available for importing from CSV and exporting to JSON: both steps can be fully automated through the CLI. The dcml-to-tilia repository demonstrates this by integrating an external corpus of musical analyses.

The desktop application

The main part of TiLiA is a cross-platform *What-You-See-Is-What-You-Get* GUI (Figure 2), designed to feel intuitive to anyone who has worked with audiovisual editing software. A TiLiA analysis consists of timelines, each containing annotations that can be aligned to an audio, video, YouTube stream or musical score. Each *timeline type* is designed to support conventional formal annotations, with the affordance to work with a variety of other analytical styles. These timelines each contain one or more *component types* (e.g., markers for marker timelines), each with different *properties* to convey information including comments, colour and labels.

We connect timelines and their components by using the loaded media as a "base layer" to which all timelines refer. Timeline components have either `time` or `start_time` and `end_time` timestamps which position them in relation to the underlying media. If a different media is loaded, the user has the option to either uniformly scale timelines to the new media or to keep them at their original position (deleting out-of-bounds components if necessary). If a beat timeline is present, the software can map timestamps to metric positions, also providing a metric-based interface for importing, exporting and interacting with timelines.

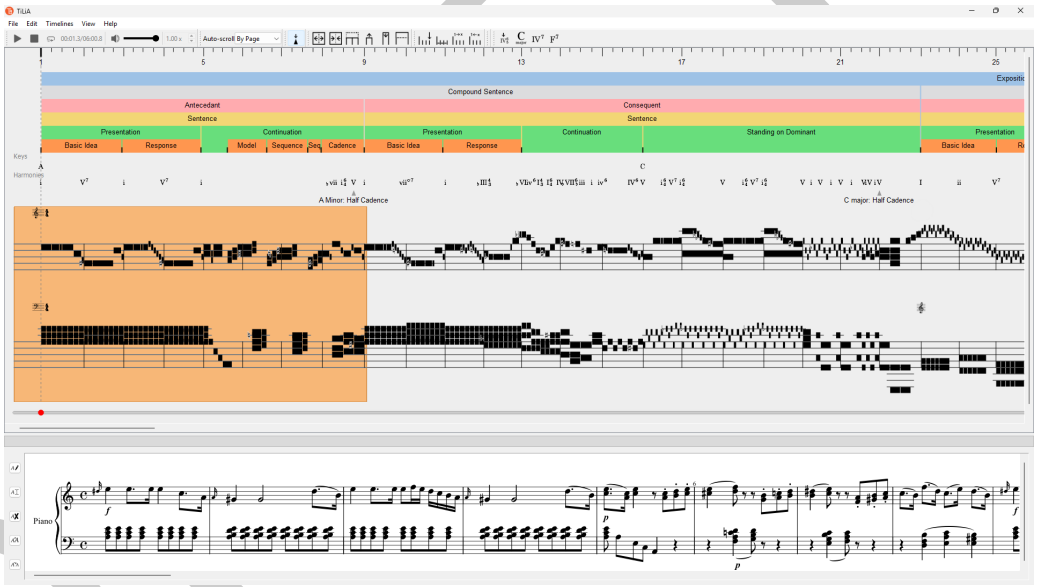


Figure 2: Excerpt of a TiLiA analysis on the desktop application.

Currently, there are eight types of timelines:

Timeline type	Purpose
Audiowave	Displaying amplitude graphs
Beat	Marking beats/measures in metrical structure, and synchronising to absolute time
Harmony	Chord symbols and roman numeral analysis
Hierarchy	Representing hierarchical relations and sequences
Marker	Annotating discrete timepoints, with optional labels
PDF	Synchronising external PDF documents with playback
Range	Representing non-hierarchical relations and sequences
Score	Synchronising digitised scores with playback

Command-line interface

In addition to the desktop GUI and the web app, a **command line interface** allows the automation of file and timeline creation and edition, media loading and external data importing. Sequences of commands can be loaded by the CLI from scripts written in plain text files, enabling programmatic conversion from various data formats to TiLiA files.

```
metadata set-media-length 120
timelines add hierarchy --name "Form"
timelines add marker --name "Formal closures"
timelines add hierarchy --name "Harmonic segments"
timelines add beat --name "Measures" --beat-pattern 1
timelines add marker --name "Harmony"
timelines add marker --name "Harm. functions"
timelines add marker --name "Keys"

load-media <path to audio> --scale-timelines yes
metadata set title "Title"
save <path to TiLiA file> --overwrite
export <path to JSON file> --overwrite
```

Online platform

The TiLiA desktop application is supported by an online platform that lets users store, share, visualise and even query existing analyses. Registration is free and is only required for uploading content. This reflects a goal of TiLiA to support not only the modernised creation and storage of analyses, but also to support the still-rare practice of analytical *collaboration* and structured *searches* on the corpora; both which benefit from network effects facilitated by the lowered entry barrier.

Outlook

The TiLiA desktop application aims to provide a user-friendly annotation tool both for MIR researchers and musicologists. It can be used to create large annotation datasets or detailed, interactive analyses of individual pieces. It can also be used in the classroom to better connect theoretical concepts with audiovisual material or as a common interface for analytical assignments.

In the future we hope to convert existing open-source datasets and integrate them into the web platform, providing a centralised hub where diversified audiovisual data can be queried and interacted with in a uniform manner. We also plan to improve audio processing and score displaying capabilities and to add more timeline types (e.g. a 2-dimensional graph timeline) to offer more diversified visualisations.

References

Balke, S., Hiemer, M., Schwab, P., Arifi-Müller, V., Meyer-Wegener, K., & Müller, M. (2017). Die Oper als Multimediaszenario – Wagners Walküren gehen online. *INFORMATIK 2017*, 75–86. https://doi.org/10.18420/in2017_04

Cannam, C., Landone, C., & Sandler, M. (2010). Sonic Visualiser: An Open Source Application for Viewing, Analysing, and Annotating Music Audio Files. *Proceedings of the 18th ACM International Conference on Multimedia*. https://doi.org/10.1145/1873951.1874248

Couprie, P. (2019). iAnalyse 5 : Le développement d'un logiciel pour la musicologie numérique. *Journées d'informatique Musicale (JIM 2019)*. https://hal.science/hal-02133538

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Fremerey, C., Kurth, F., Müller, M., & Clausen, M. (2007). A Demonstration of the SyncPlayer System. *Proceedings of the 8th International Conference on Music Information Retrieval*, 131–132. https://ismir2007.ismir.net/proceedings/ISMIR2007_p131_fremerey.pdf

Giraud, M., Groult, R., & Leguy, E. (2018). Dezrann, a Web Framework to Share Music Analysis. In S. Bhagwati & J. Bresson (Eds.), *Proceedings of the International Conference on Technologies for Music Notation and Representation – TENOR'18* (pp. 104–110). Concordia University. ISBN: 978-1-5251-0551-7

Goebl, W., & Weigl, D. M. (2024). Mei-friend: An interactive web-based editor for digital music encodings. *Journal of Open Source Software*, *9*(98), 6002. https://doi.org/10.21105/joss.06002

Guillot, P. (2025). Partiels – Exploring, Analyzing and Understanding Sounds. *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)*, 298–303. https://www.dafx.de/paper-archive/2025/DAFx25_paper_66.pdf

Hentschel, J., Neuwirth, M., & Rohrmeier, M. (2021). The Annotated Mozart Sonatas: Score, Harmony, and Cadence. *Transactions of the International Society for Music Information Retrieval*, *4*(1), 67–80. https://doi.org/10.5334/tismir.63

Isaacson, E. (2023). *Visualizing Music*. Indiana University Press. https://www.barnesandnoble.com/w/visualizing-music-eric-isaacson/1140906228?ean=9780253064738

Jarvis, B. E. (2020). *BriFormer*. Web application, version 1.2. https://www.brianedwardjarvis.com/MusicTheoryWebApps/BriFormer/briformer.html

Kurth, F., Müller, M., Damm, D., Fremerey, C., Ribbrock, A., & Clausen, M. (2005). SyncPlayer – An Advanced System for Multimodal Music Access. *Proceedings of the 6th International Conference on Music Information Retrieval*. https://ismir2005.ismir.net/proceedings/1025.pdf

Lewis, D., Page, K., & Dreyfus, L. (2021). Narratives and Exploration in a Musicology App: Supporting Scholarly Argument with the Lohengrin TimeMachine. *Proceedings of the 8th International Conference on Digital Libraries for Musicology*, 50–58. https://doi.org/10.1145/3469013.3469020

McFee, B., & Ellis, D. P. W. (2014). Analyzing song structure with spectral clustering. *Proceedings of the 15th International Society for Music Information Retrieval Conference*. https://archives.ismir.net/ismir2014/paper/000319.pdf

Müller, M., Prätzlich, T., Bohl, B., & Veit, J. (2013). Freischütz Digital: A Multimodal Scenario for Informed Music Processing. *Proceedings of the 14th International Workshop on Image Analysis for Multimedia Interactive Services (WIAMIS)*, 1–4. https://doi.org/10.1109/WIAMIS.2013.6616168

Neuwirth, M., Harasim, D., Moss, F. C., & Rohrmeier, M. (2018). The Annotated Beethoven Corpus (ABC): A Dataset of Harmonic Analyses of All Beethoven String Quartets. *Frontiers in Digital Humanities*, *5*, 16. https://doi.org/10.3389/fdigh.2018.00016

Nieto, O., McCallum, M., Davies, M. E. P., Robertson, A., Stark, A., & Egozy, E. (2019). The Harmonix Set: Beats, Downbeats, and Functional Segment Annotations of Western Popular Music. *Proceedings of the 20th International Society for Music Information Retrieval Conference*, 565–572. https://doi.org/10.5281/zenodo.3527870

Notess, M., & Swan, M. (2004). Timeliner: Building a learning tool into a digital music library. In L. Cantoni & C. McLoughlin (Eds.), *Proceedings of EdMedia + innovate learning 2004* (pp. 603–609). Association for the Advancement of Computing in Education (AACE). https://www.learntechlib.org/p/12995

Peebles, C. (2013). Using Audacity to Participate in Active Musical Listening. *Engaging Students: Essays in Music Pedagogy*. http://flipcamp.org/engagingstudents/peebles.html

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Rosenzweig, S., Scherbaum, F., Shugliashvili, D., Arifi-Müller, V., & Müller, M. (2020). Erkomaishvili Dataset: A Curated Corpus of Traditional Georgian Vocal Music for Computational Musicology. *Transactions of the International Society for Music Information Retrieval*, *3*(1), 31–41. https://doi.org/10.5334/tismir.44

Röwenstrunk, D. (2014). *Markieren, Codieren, Annotieren – Arbeiten an und mit digitalen Musikeditionen mit Edirom und MEI*.

Smith, J. B. L., Burgoyne, J. A., Fujinaga, I., De Roure, D., & Downie, J. S. (2011). Design and creation of a large-scale database of structural annotations. *Proceedings of the 12th International Society for Music Information Retrieval Conference*, 555–560.

Weigl, D. M., VanderHart, C., Rammler, D., Pescoller, M., & Goebl, W. (2023). Listen here! A web-native digital musicology environment for machine-assisted close listening. *Proceedings of the 10th International Conference on Digital Libraries for Musicology*, 109–118. https://doi.org/10.1145/3625135.3625144

Weiß, C., Arifi-Müller, V., Krause, M., Zalkow, F., Klauk, S., Kleinertz, R., & Müller, M. (2023). Wagner Ring Dataset: A Complex Opera Scenario for Music Processing and Computational Musicology. *Transactions of the International Society for Music Information Retrieval*, *6*(1), 135–149. https://doi.org/10.5334/tismir.161

Weiß, C., Zalkow, F., Arifi-Müller, V., Müller, M., Koops, H. V., Volk, A., & Grohganz, H. G. (2021). Schubert Winterreise Dataset: A Multimodal Scenario for Music Analysis. *ACM Journal on Computing and Cultural Heritage*, *14*(2), 1–18. https://doi.org/10.1145/3429743

Yorgason, B. (2018). Teaching with Timelines. *Engaging Students*, *6*. http://flipcamp.org/engagingstudents6/essays/yorgason.html

Zeitler, J., Weiß, C., Arifi-Müller, V., & Müller, M. (2024). BPSD: A Coherent Multi-Version Dataset for Analyzing the First Movements of Beethoven's Piano Sonatas. *Transactions of the International Society for Music Information Retrieval*, *7*(1), 195–212. https://doi.org/10.5334/tismir.196

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